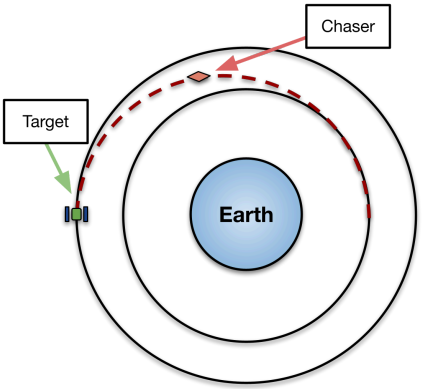
Threats in Orbit: Europe's Space Defence Gap

Diego Paniagua Silva



In April 2024, Aldoria (France), a space surveillance startup operating a network of ground-based optical tracking stations, detected a Russian satellite narrowing to within 10 km of a Norwegian communications satellite in geostationary orbit (GEO). That's close enough to intercept command signals, or even for light visual inspection.

The spacecraft in question was **Luch-2** (also known as Olymp-K-2). It was commissioned by the Russian military in 2023 as a successor to the infamous Luch-1, which was one of the first Russian satellites accused of carrying out signals intelligence (SIGINT) activities in GEO. Since its launch, Luch-2 has approached over a dozen GEO communication satellites, most notably Intelsat-39 and several Eutelsat platforms. Like its predecessor, it was judged to position itself in the narrow command link between its targets and their ground stations to collect communications data and potentially intercept command signals.

A pair of new Russian satellites, Cosmos 2589 and Cosmos 2590, launched in 2025 with similar manoeuvring capabilities to the Luch series, though their exact mission profile remains unknown. To be sure, this is orbital behaviour analysis rather than confirmed intelligence, so we should treat it with caution. However, if these satellites are indeed performing similar functions, this would represent a significant escalation in the threat to Western space infrastructure, one that is only set to intensify as Russian and Chinese spacecraft gain further experience in this kind of hybrid operation.

Detecting threats: space situational awareness



A striking feature of the Luch-2 story is that it was startups, not government agencies, that sounded the alarm. This shows the growing importance of Space Situational Awareness (SSA) companies in detecting foreign interference with strategic space assets.

Aldoria first alerted satellite operators to the movements of Luch-2 as it performed what the company described as a "sudden close approach." Slingshot Aerospace (US), a leading SSA platform, corroborated this assessment, and both companies continued to play a critical role in monitoring the Russian spacecraft. Smaller players like DigitalArsenal.io and Swiss-based S2A Systems also contributed by independently verifying Luch-2's movements. Germany-based Vyoma, developer of the Flamingo SSA constellation, represents another European entrant working to close the space surveillance gap.

Clearly, European SSA companies like Aldoria, Vyoma, and S2A Systems are at the forefront of monitoring potentially dangerous actions by state actors. But this success story also exposes a weakness in Europe's overall space defence pipeline. While these companies can track suspicious

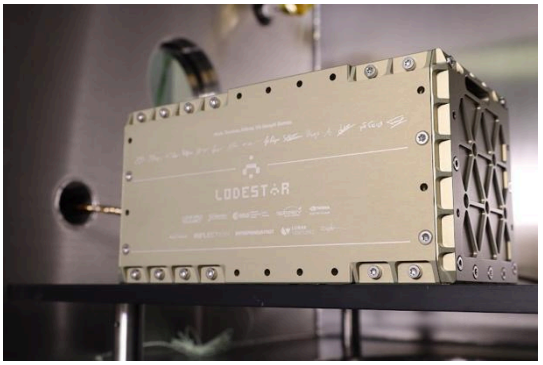
manoeuvres, they rarely have the ability to determine the purpose of such movements or understand the capabilities of the threat. Currently, they have no means of stopping any interference.

Despite knowing *where* Luch-2 is, it remains difficult to understand *what* it is doing without more information on its payload. Russia has labelled it a communications satellite, and the limited data that can be deduced from tracking (radar cross section, surface reflectivity, orbital behaviour) reveals little about the nature of its equipment. There is very little that can be done to investigate a satellite's capabilities at GEO distances without direct, close-range inspection.

The natural next step is to move from detection to threat characterisation, an entirely different challenge.

Proximity operations: from detection to characterisation

This is where companies like UK-based Lodestar Space come in. Lodestar aims to create "bodyguard" satellites that can match orbit with strategic assets and assess any threats that approach their designated target. Such close-range inspection capabilities would provide far more complete information on the architecture and intent of approaching spacecraft, while also functioning as a deterrent.



Lodestar's flagship product is **Mithril**. This innovative neural network and computer vision suite is designed to detect, classify, and respond to proximity threats. Importantly, the company has positioned Mithril as a platform-agnostic autonomy software suite, in contrast to a standalone spacecraft like many other similar startups. This is a capital-efficient approach that allows integration onto partner satellite buses. The system integrates Light Detection and Ranging (LiDAR) with machine vision and onboard AI to build a real-time picture of the space around a protected asset.

Lodestar's first test mission, TESSERACT, is set to fly aboard Exotopic's Give Me Some Space! (GMSS) mission. It was expected to launch in Q1 2026, however no launch date has been announced. This will be an "eyes and brain" test, with the aim of validating the edge-compute stack and sensor suite. This represents a crucial step in pushing Mithril from TRL 5–6 toward TRL 7. During the mission, Mithril will deploy a mounted arm with various materials attached to test its ability to image and characterise objects against different backgrounds it may encounter.

A key distinction must be made between what Lodestar is ready to demonstrate, and what is still conceptual. Mithril's sensor and compute core is believed to be ready for orbital validation, but the full bodyguard concept, including active defence capabilities, remains several development cycles away from deployment. The company is still in the process of integrating LiDAR through its ECLIPSE project, and training of the ML models against real orbital data is likely ongoing. This leads to the conclusion that Mithril is funded and in development, but not yet a demonstrated capability.

Crucially, Lodestar is not alone. US-based True Anomaly is the most advanced commercial company in this field, with its **Jackal** spacecraft carrying multi-spectral sensors for close-range detection and tracking. Jackal is complemented by **Mosaic**, a software suite similar in concept to Mithril: a hardware-agnostic AI platform for command, control, and autonomous decision-making. True

Anomaly is considerably further along, having raised over $260M and already flown prototype missions. The comparison must be made because it reveals both what is technically achievable and the scale of investment Europe currently lacks.

Responding to threats: deterrence by detection

There is a fundamental gap between understanding a threat and being protected from it. Could a bodyguard satellite physically reposition an intruding spacecraft? Could the satellite deny it optimal SIGINT geometry through close-range shadowing?

The technical, legal, and doctrinal questions are substantial, and no Western commercial entity has yet demonstrated active counter-space effects from a bodyguard platform. The Outer Space Treaty constrains but does not clearly prohibit non-destructive, reversible defensive actions, leaving a grey area that remains largely uncharted.

But it may be the case that active physical response is not the most important capability a bodyguard satellite offers. Detection itself can be a potent form of protection. SIGINT operations in GEO are effective precisely because they are deniable: the target may not know it is being surveilled, and even if suspicious activity is detected from the ground, attribution takes time and evidence is limited. A co-orbital bodyguard with high-resolution sensors changes this calculus fundamentally. It removes anonymity, eliminates deniability, and generates detailed characterisation data that can support rapid diplomatic and intelligence responses.

This deterrence-by-detection logic has parallels in other domains: a security camera doesn't physically stop a burglar, but it changes the cost-benefit calculation by ensuring the burglar will be identified. In orbital terms, a bodyguard satellite shifts the defender's posture from "we might notice weeks later via ground-based tracking" to "we are watching in real time, and we have the evidence."

The wider landscape and the European gap

This detection-response framework doesn't exist in a vacuum. The US operates the Geosynchronous Space Situational Awareness Program (GSSAP), a constellation of manoeuvrable satellites in near-GEO orbit that has been monitoring and characterising objects since 2014, an operational benchmark that Europe currently lacks. The US Space Force is also procuring its next-generation replacement, RG-XX, as a proliferated fleet of lower-cost reconnaissance satellites.



On the resilience side, the EU is investing at the infrastructure level. IRIS², a multi-orbit constellation of approximately 290 satellites, will provide encrypted communications with quantum key distribution through the EuroQCI initiative, with initial services expected by 2030. This addresses the encryption and communications resilience layer but does not address physical proximity threats.

Germany's 2025 space security strategy, meanwhile, explicitly calls for "highly agile low-signal surveillance and bodyguard satellites" and envisions €35B in space security investment between 2026 and 2030. This is one of the strongest European policy signals to date, but one that has yet to translate into hardware programmes.

The gap is clear. The US has operational surveillance constellations, flight-tested commercial protectors, and proliferated architectures in deployment. India is investing billions, with bodyguard satellite prototypes expected to fly by late 2026. Europe's concrete programmes (IRIS² and EuroQCI) address communications resilience and encryption but leave the physical protection problem largely untouched. The EU's planned "Space Shield" remains a policy concept, not an operational capability.

Closing the detection-response gap

The Luch-2 story reveals both the strength and the fragility of Europe's current position in space defence. European SSA startups proved that they could detect and track a sophisticated Russian intelligence satellite. That is a genuine achievement, one that would not have been possible five years ago. But detection without the ability to respond leaves an incomplete deterrent loop.

For European defence-tech founders, the gap between detection and response represents the defining challenge and opportunity of the next decade. The technical path forward is becoming clearer: from ground-based tracking to space-based situational awareness, to proximity characterisation, and eventually to active defence. Each step up this ladder requires not just new technology but new doctrine, new legal frameworks, and new forms of government-startup collaboration.

Lodestar's Mithril, if its early 2026 validation mission succeeds, will be a meaningful proof point, one that demonstrates a small European startup can field proximity sensing capabilities that, until recently, only major state programmes could attempt. True Anomaly's progress in the US shows what's possible with greater scale and funding. Germany's ambitious spending commitments suggest the political will is emerging.

But the window is narrowing. Russia is not standing still: the Cosmos 2589/2590 pair suggests an expanding fleet of orbital surveillance platforms, and China's rapid advances in co-orbital capabilities add further urgency. Every month that passes without European response capabilities in orbit is a month in which adversaries refine their ability to operate with impunity near Europe's most critical space assets.

The question facing European space defence is no longer whether the threat is real. Luch-2 settled that debate. The question is whether we can close the gap between knowing what is happening in our own orbital neighbourhood and being able to do something about it.